

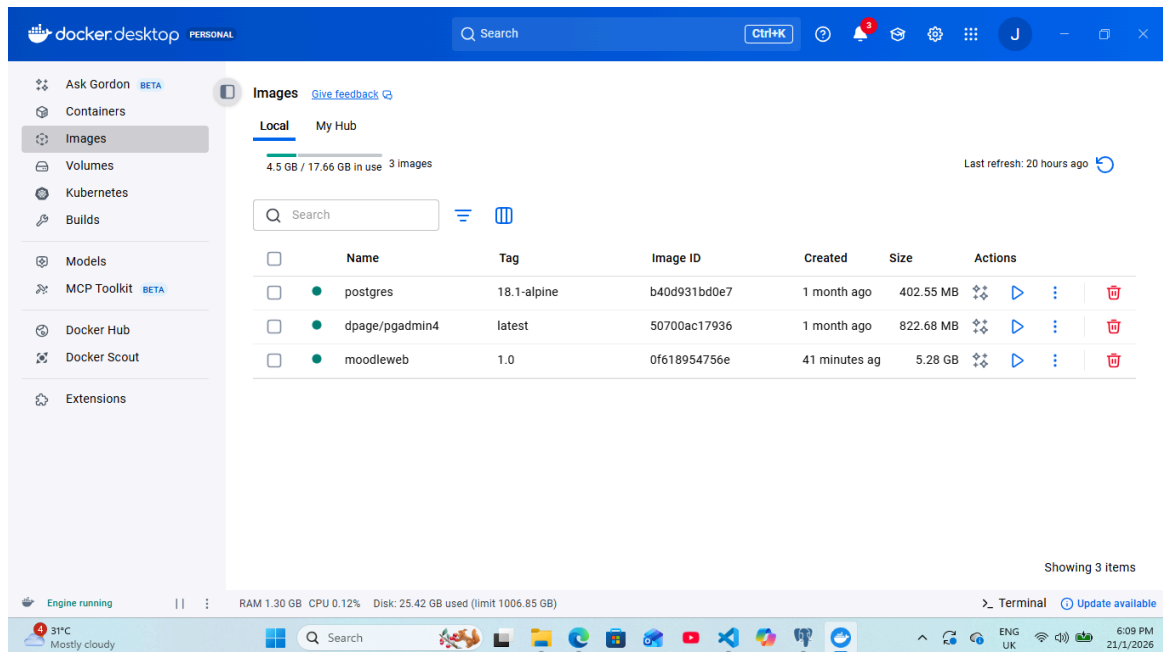
WIF3005: Software Maintenance and Evolution

Name: Joanne Lim Zi Xuan

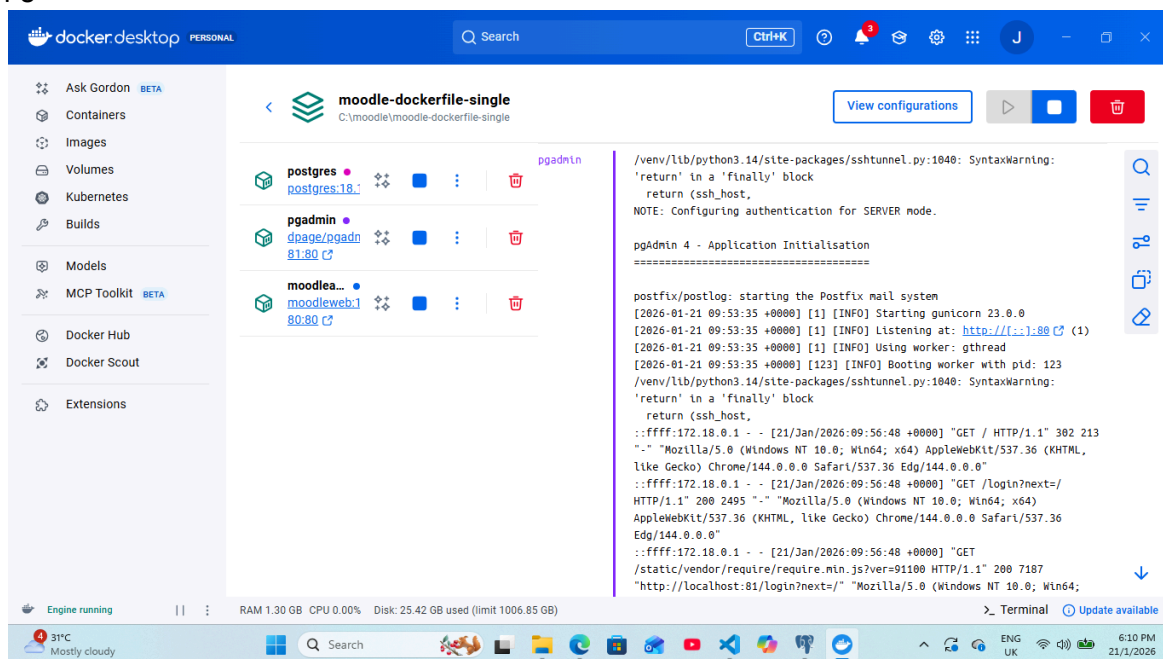
Matric no.: 22004882/1

Build

The “moodleweb” image is the image that I built.



The Moodle web application requires external services to run, including PostgresDB and pgadmin.



docker.desktop PERSONAL

Search Ctrl+K

Ask Gordon **BETA**

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit **BETA**

Docker Hub

Docker Scout

Extensions

moodle-dockerfile-single
C:\moodle\moodle-dockerfile-single

View configurations

postgres:18.1

pgadmin:deage/pgadmin:81.80

moodleadmin:moodleweb:80.80

postgres

moodleapp

Edg/144.0.0.0
::ffff:172.18.0.1 - - [21/Jan/2026:09:56:49 +0000] "GET /static/js/generated/fonts/Roboto-Regular..ttf HTTP/1.1" 200 0 "http://localhost:81/static/js/generated/style.css?ver=91100" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"

2026-01-21 09:57:31.353 UTC [64] LOG: checkpoint starting: ttime
2026-01-21 09:57:35.000 UTC [64] LOG: checkpoint complete: wrote 36 buffers (0.2%), wrote 3 SLRU buffers; 0 WAL file(s) added, 0 removed, 0 recycled; write=3.620 s, sync=0.009 s, total=3.648 s; sync files=12, longest=0.004 s, average=0.001 s; distance=248 kB, estimate=248 kB; lsn=0/1BDD3E0, redo lsn=0/1BDD3B8

172.18.0.1 - - [21/Jan/2026:09:57:39 +0000] "-" 408 0 "-" "-"
172.18.0.1 - - [21/Jan/2026:10:04:35 +0000] "GET / HTTP/1.1" 302 254 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"
172.18.0.1 - - [21/Jan/2026:10:04:35 +0000] "GET /install.php HTTP/1.1" 200 5198 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"
172.18.0.1 - - [21/Jan/2026:10:04:39 +0000] "GET /pix/moodleLogo.png HTTP/1.1" 200 17581 "http://localhost/install.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"
172.18.0.1 - - [21/Jan/2026:10:04:39 +0000] "GET /install/css.php HTTP/1.1" 200 141266 "http://localhost/install.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"

Engine running

RAM 1.30 GB CPU 0.12% Disk: 25.42 GB used (limit 1006.85 GB)

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Search

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6:10 PM 21/1/2026

docker.desktop PERSONAL

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172.18.0.1 - - [21/Jan/2026:09:57:39 +0000] "-" 408 0 "-" "-"
172.18.0.1 - - [21/Jan/2026:10:04:35 +0000] "GET / HTTP/1.1" 302 254 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"
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172.18.0.1 - - [21/Jan/2026:10:04:39 +0000] "GET /theme/boost/pix/favicon.ico HTTP/1.1" 200 1450 "http://localhost/install.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 Edg/144.0.0.0"

Engine running

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6:09 PM 21/1/2026

Test

pgAdmin 4 x Installation | Moodle 5.0.4+ (Build: X +

localhost/install.php

Spotify Investopedia Gmail YouTube Google Docs SPeCTRUMv4 Moodle Plugins dir... Sean Prashad Leet... TUM Moodle TUMonline > Other favourites

Installation

Language

Choose a language

Please choose a language for the installation. This language will also be used as the default language for the site, though it may be changed later.

Language

Next »



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pgAdmin 4 x Installation | Moodle 5.0.4+ (Build: X +

localhost/install.php Summarise ☆ > Other favourites

Spotify Investopedia Gmail YouTube Google Docs SPeCTRUMv4 Moodle Plugins dir... Sean Prashad Leet... TUM Moodle TUMonline

Moodle directory
The full path to the directory containing the Moodle code.

Data directory
A directory where Moodle will store all file content uploaded by users.
This directory should be both readable and writeable by the web server user (usually 'www-data', 'nobody', or 'apache').
It must not be directly accessible over the web.
If the directory does not currently exist, the installation process will attempt to create it.

Web address

Moodle directory

Data directory

« Previous Next »



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pgAdmin 4

Installation | Moodle 5.0.4+ (Build: X

localhost/install.php

Summarise

Spotify Investopedia Gmail YouTube Google Docs SPeCTRUMv4 Moodle Plugins dir... Sean Prashad Leet... TUM Moodle TUMonline Other favourites

Installation

Database

Choose database driver

Moodle supports several types of database servers. Please contact server administrator if you do not know which type to use.

Type

[< Previous](#)[Next >](#)



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Search

ENG UK 6:11 PM 21/1/2026

Dashboard | SME pgAdmin 4 github - Search

localhost/my/ Summarise A Chat

Spotify Investopedia Gmail YouTube Google Docs SPeCTRUMv4 Moodle Plugins dir... Sean Prashad Leet... TUM Moodle TUMonline Other favourites

SME Home Dashboard My courses Site administration AU Edit mode

Dashboard

Welcome, Admin! 🙌

Timeline

Next 7 days ▾ Sort by dates ▾ Search by activity type or name



No in-progress courses

Calendar

All courses ▾

New event

?

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localhost/?redirect=0 A Chat

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SME Home Dashboard My courses Site administration AU Edit mode

SME

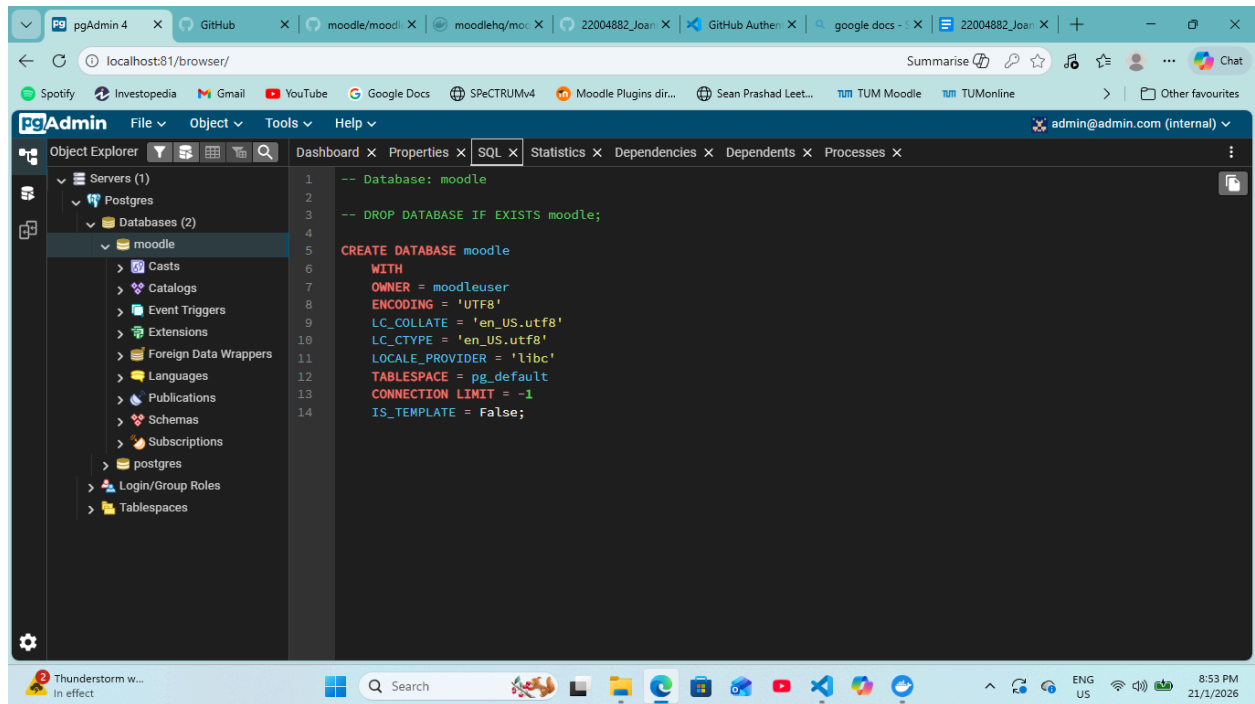
Home Settings Participants Reports Question banks More ▾

Available courses

Add a new course

?

Thunderstorm w... In effect Search ENG US 7:40 PM 21/1/2026



Build Without Database

```
File Edit Selection View Go ... Search
Welcome dockerfile X
C:\Users\joann> moodle2> dockerfile > ...
14 git checkout MOODLE_500_STABLE && \
15 cp -rf ./.* /var/www/html/
16
17 # Set PHP settings for Moodle: max_input_vars and OPcache
18 RUN echo "max_input_vars=5000" >> /usr/local/etc/php/conf.d/docker-php-moodle.ini && \
19 echo "opcache.enable=1" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \
20 echo "opcache.enable_cli=1" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \
21 echo "opcache.memory_consumption=128" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\joann\moodle2> docker build -t mymoodle .
[+] Building 3.3s (11/11) FINISHED
=> [internal] load build definition from dockerfile
=> [internal] load metadata for docker.io/library/php:8.2.4-apache
=> [internal] load .dockerignore
=> [internal] transfer context: 28
=> [1/7] FROM docker.io/library/php:8.2.4-apache@sha256:259e142a3e842c7d287ae2098ca68ba3efecb77a5f1b52af3a1cda3cd1dd17
=> resolve docker.io/library/php:8.2.4-apache@sha256:259e142a3e842c7d287ae2098ca68ba3efecb77a5f1b52af3a1cda3cd1dd17
=> CACHED [2/7] RUN apt-get update && apt-get install -y --no-install-recommends unzip git curl libzip-dev libjpeg-dev libfreetype6-
=> CACHED [3/7] RUN echo "max_input_vars=5000" >> /usr/local/etc/php/conf.d/docker-php-moodle.ini && echo "opcache.enable=1" >> /usr/local/etc/php/
=> CACHED [4/7] RUN mkdir -p /var/www/moodledata
=> CACHED [5/7] RUN echo " * * * * www-data /usr/local/bin/php /var/www/html/admin/cli/cron.php > /dev/null" > /etc/cron.d/moodle-cron && chmod 06
=> CACHED [6/7] WORKDIR /var/www/html
=> CACHED [7/7] RUN chown -R www-data:www-data /var/www/ && chmod -R 755 /var/www
=> exporting to image
=> exporting layers
=> exporting manifest sha256:e93cad652fd1e6fdac7e013b9673c678f5071f03e23699516705f9d5d488905f
=> exporting config sha256:15a4795e7890b03ce880dd0c8e9536bd32c43e26bb0191ce4b0feb1cdef4f4b6
=> exporting attestation manifest sha256:b5836ded3e5137bae0026554869dea153e92fcb95de614d253ccb57e35114490
=> exporting manifest list sha256:330b3211a4574a2039cb46b8450b8ae25763ae97d6b583aa8b348a9c34a5194d
=> naming to docker.io/library/mymoodle:latest
=> unpacking to docker.io/library/mymoodle:latest

Ln 41, Col 10 Spaces: 4 UTF-8 CRLF {} Dockerfile
31°C Mostly cloudy 2:42 PM 22/1/2026
```

```
File Edit Selection View Go ... Search
Welcome dockerfile X
C:\Users\joann> moodle2> dockerfile > ...
14 git checkout MOODLE_500_STABLE && \
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19 echo "opcache.enable=1" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \
20 echo "opcache.enable_cli=1" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \
21 echo "opcache.memory_consumption=128" >> /usr/local/etc/php/conf.d/docker-php-opcache.ini && \

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
=> CACHED [7/7] RUN chown -R www-data:www-data /var/www/ && chmod -R 755 /var/www
=> exporting to image
=> exporting layers
=> exporting manifest sha256:e93cad652fd1e6fdac7e013b9673c678f5071f03e23699516705f9d5d488905f
=> exporting config sha256:15a4795e7890b03ce880dd0c8e9536bd32c43e26bb0191ce4b0feb1cdef4f4b6
=> exporting attestation manifest sha256:b5836ded3e5137bae0026554869dea153e92fcb95de614d253ccb57e35114490
=> exporting manifest list sha256:330b3211a4574a2039cb46b8450b8ae25763ae97d6b583aa8b348a9c34a5194d
=> naming to docker.io/library/mymoodle:latest
=> unpacking to docker.io/library/mymoodle:latest

View build details: docker-desktop:///dashboard/build/desktop-linux/desktop-linux/s6tw/ecp7b3vaooqqqfhh10h
PS C:\Users\joann\moodle2> docker run -p 5000:5000 mymoodle
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress
this message
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress
this message
[Thu Jan 22 06:43:39.710283 2026] [mpm_prefork:notice] [pid 1] AH00163: Apache/2.4.56 (Debian) PHP/8.2.4 configured -- resuming normal operations
[Thu Jan 22 06:43:39.710447 2026] [core:notice] [pid 1] AH00094: Command line: 'apache2 -D FOREGROUND'

Ln 41, Col 10 Spaces: 4 UTF-8 CRLF {} Dockerfile
31°C Mostly cloudy 2:48 PM 22/1/2026
```

